ARCANUM: Zero-Plaintext Confidential Compilation via Air-Gapped Zero-Knowledge Proving (AG-ZKP)

Verifiable Software Supply-Chain Integrity & Sovereign IP Protection for B2G, Defense, and Institutional Infrastructure

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EXECUTIVE SUMMARY

Modern software compilation and cryptographic proving pipelines suffer from a fatal architectural paradox: to verify that a compiled binary or execution trace conforms to strict security policies, the underlying source code and secret witness w must be materialized in plaintext inside the volatile RAM of a network-connected host. For defense contractors, sovereign government agencies (B2G), and institutional financial infrastructures, exposing mission-critical source code or cryptographic witnesses to cloud hypervisors, operating-system kernels, or microarchitectural TEE side channels is unacceptable. **Arcanum** resolves this paradox by unifying **Zero-Plaintext Confidential Compilation** with **Air-Gapped Zero-Knowledge Proving (AG-ZKP)**. By housing the compiler and zkVM prover inside a physically isolated, network-stripped hardware boundary (P_{AG}) connected solely via unidirectional optical or data-diode links, Arcanum proves that a binary derives faithfully from audited source code without a single line of plaintext source or witness data ever touching an online operating system.

1. THE SOVEREIGN COMPILATION & PLAINTEXT RAM CRISIS

Historically, protecting critical software intellectual property relied on shipping stripped binaries without source code —a paradigm analyzed in classic binary translation literature [8] that has collapsed under modern LLM-assisted decompilation and supply-chain attacks. Conversely, when sovereign regulators or enterprise auditors demand proof that an executable binary is free from backdoors and matches a certified source commit, traditional CI/CD pipelines require full source disclosure inside online build servers. Even when compilation or ZK proof generation is wrapped inside Trusted Execution Environments (Intel SGX/TDX, AMD SEV-SNP), non-uniform memory access patterns during Abstract Syntax Tree (AST) lowering, Multi-Scalar Multiplications (MSM), and Number Theoretic Transforms (NTT) leak secret witness state across enclave boundaries.

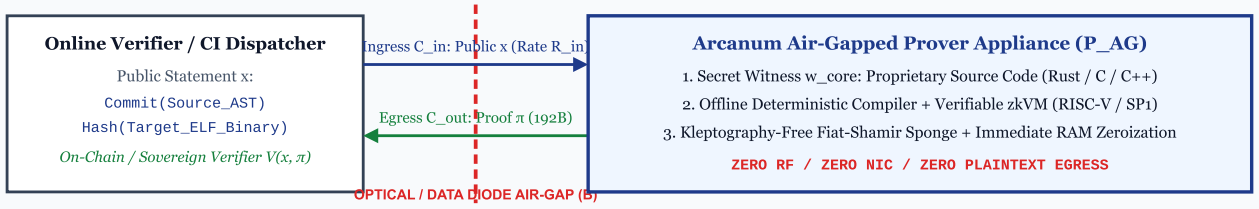


Figure 1: Arcanum architecture decoupling proprietary source compilation inside P_AG from online verification across boundary B.

Arcanum eliminates this attack surface by treating proprietary source code S and secret build parameters as a persistent offline witness w_{core} locked permanently inside an air-gapped proving appliance P_{AG}. Only the succinct cryptographic proof π crosses the unidirectional optical/diode boundary.

2. CRYPTOGRAPHIC CORE: AG-ZKP VS. FHE & SUBLIMINAL KLEPTOGRAPHY

In our companion cryptographic research [1], we evaluate why purely mathematical alternatives such as Fully Homomorphic Encryption (*FHE + ZKP* blind proving) fail in production compilation pipelines: evaluating prime-field SNARK/STARK circuits ($\mathbb{F}_p$) inside RLWE polynomial rings triggers continuous FHE bootstrapping ($10^4\times$–$10^6\times$ slowdown) and $>10^3\times$ RAM expansion. Conversely, Arcanum's **AG-ZKP** engine runs at $1\times$ native hardware speed inside P_{AG} while neutralizing supply-chain compiler trojans (*ZK Kleptography*) by enforcing **Deterministic Fiat-Shamir Blinding**:

$$r_i \leftarrow H_{FS}(\texttt{"ARCANUM-AG-ZKP-V1"} \,\|\, vk_{hash} \,\|\, x \,\|\, H(w_{core}) \,\|\, \sigma_{hw} \,\|\, i) \bmod p \implies T_{total} = (|x| / R_{in}) + T_{zkVM}(N) + (|\pi| / R_{out})$$

Compilation & Proving Paradigm	Source / Witness RAM Exposure	Proving Overhead	Kleptographic / Exfiltration Risk
Standard Cloud CI/CD + Online Prover	100% Plaintext in Host Kernel RAM	$1\times$ Baseline	Critical (Full TCP/IP & OS access)
Hardware TEE Enclave (SGX / SEV-SNP)	Encrypted DRAM; side-channel leaky	$1.3\times$–$2.5\times$	High (Cache-timing & host NIC relay)
MPC Split Proving Across Nodes	Secret-shared across k online hosts	$5\times$–$25\times$	Moderate (Requires active network links)
Homomorphic Blind Proving (FHE+ZKP)	Encrypted ciphertexts ($>10^3\times$ RAM)	$10^4\times$–$10^6\times$	Requires isolated decryption authority
Arcanum (AG-ZKP Offline Appliance)	**Zero Online RAM (Isolated in P_AG)**	**$1\times$ Native + <0.1s I/O**	**Eliminated (Unidirectional diode + Det-FS)**

3. B2G, DEFENSE & INSTITUTIONAL DEPLOYMENT VERTICALS

1. Sovereign Defense & SCIF Compliance (B2G)

Allows defense agencies and aerospace contractors to prove that mission-critical firmware compiled inside classified SCIF / air-gapped facilities matches audited safety specifications and contains zero unauthorized patches—emitting only a 192-byte proof across fiber data diodes.

2. Critical Infrastructure & Energy Grids

Enables nuclear, telecom, and power-grid operators to cryptographically attest software supply-chain integrity and deterministic binary reproducibility to national cybersecurity regulators without connecting industrial control networks to the internet.

3. Institutional Finance & Arbitrum Stylus L2

Quantitative funds and RWA institutions compile proprietary Rust/C++ risk engines offline inside Arcanum and verify execution proofs directly on-chain via **Arbitrum Stylus** WASM verifiers—achieving regulatory compliance with zero algorithmic IP leakage.

4. Sovereign AI & Proprietary Model Weights

Protects billion-dollar foundational AI weights and defense inference models inside offline hardware appliances while proving to external clients that outputs were generated by the authentic, unmodified model commitment.

4. PHASED IMPLEMENTATION ROADMAP & REFERENCES

Phase 1 (Core Rust CLI & Deterministic zkVM Bridge): Offline Rust compilation wrapper integrated with Succinct SP1 / RISC-V deterministic execution and Poseidon2 Fiat-Shamir blinding. **Phase 2 (Optical & Data-Diode Transport):** High-throughput unidirectional framing over QR optical streams and hardware fiber diodes with automated volatile RAM zeroization. **Phase 3 (Sovereign & L2 Verifiers):** Native verification contracts for Arbitrum Stylus, Starknet, and air-gapped B2G audit terminals.

1. **[AK26]** M. Atakan and A. Khatri, "Air-Gapped Zero-Knowledge Proving (AG-ZKP): Formalizing Witness Isolation, Unidirectional I/O Bounds, and Kleptographic Resistance in Offline Provers," *Zenodo / CERN Preprint v1.1*, Oct. 2026. `github.com/Sidistr/airgapped-zk-paper`
2. **[Groth16]** J. Groth, "On the Size of Pairing-Based Non-interactive Arguments," *EUROCRYPT 2016*, LNCS 9666, pp. 305–326.